

statement is the proposition "It is not the case that there is a student in this class who has taken a course in calculus." This is equivalent to "Every student in this class has not taken calculus." which is just the universal quantification of the negation of the original propositional function, or, phrased in the language of quantifiers,  $\boxed{\forall x \neg Q(x)}$ .

This example illustrates the equivalence  $\boxed{\neg \exists x Q(x) \equiv \forall x \neg Q(x)}$

To show that  $\neg \exists x Q(x)$  and  $\forall x \neg Q(x)$  are logically equivalent no matter what  $Q(x)$  is and what the domain is, first note that  $\neg \exists x Q(x)$  is true if and only if  $\exists x Q(x)$  is false. This is true if and only if no  $x$  exists in the domain for which  $Q(x)$  is true. Next, note that no  $x$  exists in the domain for which  $Q(x)$  is true if and only if  $Q(x)$  is false for every  $x$  in the domain. Finally, note that  $Q(x)$  is false for every  $x$  in the domain if and only if  $\neg Q(x)$  is true for all  $x$  in the domain, which holds if and only if  $\forall x \neg Q(x)$  is true. Putting these steps together, we see that  $\neg \exists x Q(x)$  is true if and only if  $\forall x \neg Q(x)$  is true. We conclude that  $\neg \exists x Q(x)$  and  $\forall x \neg Q(x)$  are logically equivalent.

The rules for negations for quantifiers are called De Morgan's laws for quantifiers.

Remark: When the domain of a predicate  $P(x)$  consists of  $n$  elements, where  $n$  is a positive integer greater than one, the rules for negating quantified statements are exactly the same as De Morgan's laws discussed in Section 1.5. This is why these rules are called De Morgan's laws for quantifiers. When the domain has  $n$  elements  $x_1, x_2, \dots, x_n$ , it follows that  $\forall x P(x)$  is the same as  $\neg (P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n))$ , which is equivalent to  $\neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n)$  by De Morgan's law, and this is the same as  $\exists x \neg P(x)$ . Similarly,  $\neg \exists x P(x)$  is the same as  $\neg (P(x_1) \vee P(x_2) \vee \dots \vee P(x_n))$ , which by De Morgan's law is equivalent to  $\neg P(x_1) \wedge \neg P(x_2) \wedge \dots \wedge \neg P(x_n)$ , and this is the same as  $\forall x \neg P(x)$ .